

OCR Further Mathematics
Statistics - Non-parametric Tests
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) In a school opinion poll a random sample of 8 pupils were asked to rate school lunches on a scale of 0 to 20. The results were as follows.

0 1 2 3 4 10 11 13

After a new menu was introduced, the test was repeated with a different random sample of 8 pupils. The results were as follows.

7 8 9 14 15 17 19 20

Carry out an appropriate Wilcoxon test at the 5% significance level to test whether pupils' opinions of school lunches have changed. [8]

- (b) A statistics student tells the organisers of the opinion poll that it would have been better to have asked the same 8 pupils both times.

Explain why the statistics student's suggestion would produce a better test. [1]

- (c) State which test should be used if the student's suggestion is followed. [1]

- (d) You are given that there are 12 870 ways in which 8 different integers can be chosen from the integers 1 to 16 inclusive.

Estimate the number of ways of selecting 8 different digits between 1 and 16 inclusive that have a sum less than or equal to the critical value used in the test in the first part of this question. [2]

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- 2(a) Jo can use either of two different routes, A or B, for her journey to school. She believes that route A has shorter journey times. She measures how long her journey takes for 17 journeys by route A and 12 journeys by route B. She ranks the 29 journeys in increasing order of time taken, and she finds that the sum of the ranks of the journeys by route B is 219.

Test at the 10% significance level whether route A has shorter journey times than route B.

[8]



- (b) State an assumption about the 29 journeys which is necessary for the conclusion of the test to be valid.

[1]

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- 3(a) After a holiday organised for a group, the company organising the holiday obtained scores out of 10 for six different aspects of the holiday. The company obtained responses from 100 couples and 100 single travellers. The total scores for each of the aspects are given in the following table.

Aspect	Couples	Single travellers
Organisation	884	867
Travel	710	633
Food	692	675
Leader	898	898
Included visits	561	736
Optional visits	683	712

Fred wishes to test whether there is significant positive correlation between the scores given by the two categories.

Explain why it is probably not appropriate to use Pearson's product-moment correlation coefficient.

[1]

(b) Carry out an appropriate test at the 1% level.

[7]



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(c) Explain what is meant by the statement that the test carried out in the part above is a non-parametric test. [1]

- 4 A university course was taught by two different professors. Students could choose whether to attend the lectures given by Professor Q or the lectures given by Professor R . At the end of the course all the students took the same examination.

The examination marks of a random sample of 30 students taught by Professor Q and a random sample of 24 students taught by Professor R were ranked. The sum of the ranks of the students taught by Professor Q was 726.

Test at the 5% significance level whether there is a difference in the ranks of the students taught by the two professors. [10]



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- 5 The reaction times, in milliseconds, of all adult males in a standard experiment have a symmetrical distribution with mean and median both equal to 700 and standard deviation 125.

The reaction times of a random sample of 6 international athletes are measured and the results are as follows:

702 631 540 714 575 480

It is required to test whether international athletes have a mean reaction time which is less than 700.

- (a) Assume first that the reaction times of international athletes have the distribution $N(\mu, 125^2)$.

Test at the 5% significance level whether $\mu < 700$.

[7]



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- (b) Now assume only that the distribution of the data is symmetrical, but not necessarily normal.

- (i) State with a reason why a Wilcoxon test is preferable to a sign test.

[1]

- (ii) Use an appropriate Wilcoxon test at the 5% significance level to test whether the median reaction time of international athletes is less than 700.

[6]



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- (c) Explain why the significance tests in part (a) and part (b)(ii) could produce different results.

[1]

- 6 In an experiment to investigate the effect of background music in carrying out work, ten students were each given a task. Five of the students did the task in silence and the other five did the task with background music. The scores on the tasks were as follows.

Silence	43	46	55	58	61
Background music	19	31	38	52	70

- (a) Use a Wilcoxon rank-sum test to test at the 10% level whether the presence of background music affects scores.

[7]



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- (b) A statistician suggests that the experiment is redesigned so that each student takes one task in silence and another task with background music. The differences in the test scores would then be analysed using a paired-sample method. State an advantage in redesigning the experiment in this way.

[1]

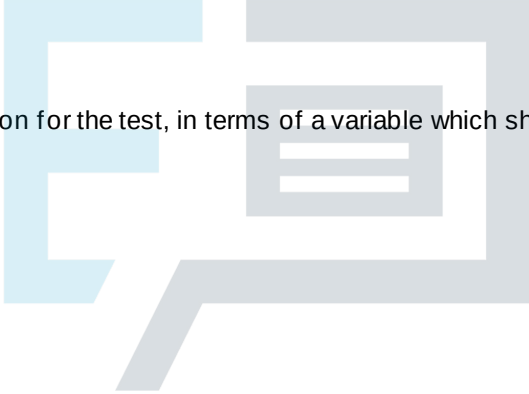
- 7 Sheena travels to school by bus. She records the number of minutes, T , that her bus is late on each of 32 days. She believes that on average T is greater than 5, and she carries out a significance test at the 5% level.

(a) State a condition needed for a Wilcoxon test to be valid in this case. [1]

Assume now that this condition is satisfied.

(b) State an advantage of using a Wilcoxon test rather than a sign test. [1]

(c) Calculate the critical region for the test, in terms of a variable which should be defined. [7]



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8 The heights of eleven randomly selected primary school children are measured. The results, in metres, are

Girls	1.48	1.31	1.63	1.38	1.56	1.57
Boys	1.44	1.35	1.32	1.28	1.27.	

- (i) Use a Wilcoxon rank-sum test, at the 1% significance level, to test whether primary school girls are taller than primary school boys. [6]



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- (ii) It is decided to repeat the test, using larger random samples. The heights of twenty girls and eighteen boys are measured. Find the greatest value of the test statistic W which will result in the conclusion that there is evidence, at the 1% level of significance, that primary school girls are taller than primary school boys.

[6]



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- 9 A meteorologist claims that the median daily rainfall in London is 2.2 mm. A single sample sign test is to be used to test the claim, using the following hypotheses:

H_0 : a sample comes from a population with median 2.2,

H_1 : the sample does not come from a population with median 2.2.

30 randomly selected observations of daily rainfall in London are compared with 2.2, and given a '+' sign if greater than 2.2 and a '-' sign if less than 2.2. (You may assume that no data values are exactly equal to 2.2.) The test is to be carried out at the 5% level of significance. Let the number of '+' signs be k . Find, in terms of k , the critical region for the test showing the values of any relevant probabilities.

[4]



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- 10 Ten randomly selected joints of meat, A, B, ..., J, are each cut in half. One half of each joint of meat is frozen and packed using process 1 and the other half is frozen and packed using a new process 2. All the halves are then placed in the same freezer. For each pack, the number of days that the meat takes to spoil is found. The results are shown in the table.

Joint	A	B	C	D	E	F	G	H	I	J
Process 1	78	108	80	171	184	153	50	145	91	156
Process 2	111	106	77	196	230	148	58	155	90	170

Carry out a paired sample t -test, at the 5% significance level, to investigate whether meat frozen and wrapped using process 2 takes longer to spoil than meat frozen and wrapped using process 1. State any necessary assumption.

[10]



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- 11 A psychologist investigated the scores of pairs of twins on an aptitude test. Seven pairs of twins were chosen randomly, and the scores are given in the following table.

Elder twin	65	37	60	79	39	40	88
Younger twin	58	39	61	62	50	26	84

- (a) Carry out an appropriate Wilcoxon test at the 10% significance level to investigate whether there is evidence of a difference in test scores between the elder and the younger of a pair of twins.

[6]



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- (b) Explain the advantage in this case of a Wilcoxon test over a sign test.

[1]

- 12 The manufacturer of a painkiller, designed to relieve headaches, claims that people taking the painkiller feel relief in at most 30 minutes, on average. A random sample of eight users of the painkiller recorded the times it took for them to feel relief from their headaches. These times, in minutes, were as follows:

33 39 29 35 40 32 26 37

Use a Wilcoxon single-sample signed-rank test at the 5% significance level to test the manufacturer's claim, stating a necessary assumption.

[8]

- 13 In a two-tail Wilcoxon rank-sum test, the sample sizes are 13 and 15. The sum of the ranks for the sample of size 13 is 135. Carry out the test at the 5% level of significance.

[9]

- 14 A teacher believes that the calculator paper in a GCSE Mathematics examination was easier than the non-calculator paper. The marks of a random sample of ten students are shown in the table.

Student	A	B	C	D	E	F	G	H	I	J
Mark on paper 1 (non-calculator)	66	79	58	87	67	55	75	62	50	84
Mark on paper 2 (calculator)	57	84	70	90	75	42	82	72	65	82

- (i) Use a Wilcoxon signed-rank test, at the 5% significance level, to test the teacher's belief.

[7]

- (ii) State the assumption necessary for this test to be applied.

[1]

15 A Wilcoxon rank-sum test with samples of sizes 11 and 12 is carried out.

(i) What is the least possible value of the test statistic W ?

[2]



(ii) The null hypothesis is that the two samples came from identical populations. Given that the null hypothesis was rejected at the 1% level using a 2-tail test, find the set of possible values of W .

[6]

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- 16 Two drugs, I and II, for alleviating hay fever are trialled in a hospital on each of 12 volunteer patients. Each received drug I on one day and drug II on a different day. After receiving a drug, the number of times each patient sneezed over a period of one hour was noted. The results are given in the table.

Patient	1	2	3	4	5	6	7	8	9	10	11	12
Drug I	11	34	19	16	10	29	6	17	20	13	4	25
Drug II	12	20	10	18	3	21	9	13	10	19	9	12

The patients may be considered to be a random sample of all hay fever sufferers.
A researcher believes that patients taking drug II sneeze less than patients taking drug I.

Test this belief using the Wilcoxon signed rank test at the 5% significance level.

[7]



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- 17 The effect of water salinity on the growth of a type of grass was studied by a biologist. A random sample of 22 seedlings was divided into two groups *A* and *B*, each of size 11.

Group *A* was treated with water of 0% salinity and group *B* was treated with water of 0.5% salinity. After three weeks the height (in cm) of each seedling was measured with the following results, which are ordered for convenience.

Group <i>A</i>	8.6	9.4	9.7	9.8	10.1	10.5	11.0	11.2	11.8	12.7	12.9
Group <i>B</i>	7.4	8.4	8.5	8.8	9.2	9.3	9.5	9.9	10.0	11.1	11.3

Jeffery was asked to test whether the two treatments resulted, on average, in a difference in growth. He chose the Wilcoxon rank sum test.

- (i) Justify Jeffery's choice of test.

[1]

- (ii) Carry out the test at the 5% significance level.

[9]

- 18 Ten archers shot at targets with two types of bow. Their scores out of 100 are shown in the table.

Archer	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
Bow type <i>P</i>	95	97	92	85	87	92	90	89	98	77
Bow type <i>Q</i>	91	91	88	90	80	88	93	85	94	84

- (i) Use the sign test, at the 5% level of significance, to test the hypothesis that bow type *P* is better than bow type *Q*.

[7]

- (ii) Why would a Wilcoxon signed rank test, if valid, be a better test than the sign test?

[1]

- 19 Low density lipoprotein (LDL) cholesterol is known as 'bad' cholesterol. 15 randomly chosen patients, each with an LDL level of 190 mg per decilitre of blood, were given one of two treatments, chosen at random. After twelve weeks their LDL levels, in mg per decilitre, were as follows.

Treatment A	189	168	176	186	183	187	188	
Treatment B	177	179	173	180	178	170	175	174

Use a Wilcoxon rank sum test, at the 5% level of significance, to test whether the LDL levels of patients given treatment B are lower than the LDL levels of patients given treatment A.

[8]



END OF QUESTION PAPER

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